INCH–POUND

MIL–F–51194C(EA)
AMENDMENT 2
28 July 2000

SUPERSEDING
AMENDMENT 1
7 June 1991

MILITARY SPECIFICATION

FILTER, PARTICULATE, 20 CFM, M19

Inactive for new design after 28 September 1995

This amendment forms a part of MIL–F–51194C(EA), dated 6 September 1990, and is approved for use by the U.S. Army Edgewood Chemical Biological Center, Department of the Army aand is available for use by all Departments and Agencies of the Department of Defense.

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2.1.2: Delete the following: "B136–42–850–2 – Penetrometer, Filter Testing, DOP, Q76 Assembly

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3.3: Delete entire paragraph and substitute the following:

"3.3 Resistance to airflow. The resistance to airflow shall be no greater than 1.25 inches of water gage (iwg) at an airflow rate of 20 standard cubic feet per minute (scfm) when tested as specified in 4.4.6.1."

3.4: Delete entire paragraph and substitute the following:

"3.4 Filtration efficiency. The filters shall be no less than 99.97% efficient in the removal of aerosols with a mean diameter of 0.3 microns when tested in accordance with 4.4.6.2."

AMSC N/A FSC 4240

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3.5:
Line 1: Delete "...DOP smoke penetration..." and substitute "...filtration efficiency..."
Line 3: Delete the following: (19 millimeters)
Line 4: Delete "...4.4.6.2." and substitute "...4.4.6.3"

3.7:
Line 1: Delete "15 cubic centimeters..." and substitute "91 cubic inches..."
Line 2: Delete "...4.4.6.3" and substitute "...4.4.6.4"

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4.4.5(a), second critical characteristic:
2nd column: Delete "DOP smoke penetration" and substitute "Filtration efficiency"
5th column: Delete "4.4.6.1" and substitute "4.4.6.2"

4.4.5(a), first two major characteristics:
5th column: Delete "4.4.6.2" and substitute "4.4.6.3**"
5th column: Delete "4.4.6.3" and substitute "4.4.6.4"

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4.4.6.1: Delete entire paragraph and substitute the following:
"4.4.6.1 Resistance to airflow. The filter shall be tested for airflow resistance by passing an airflow of at least 20 scfm through the filter at a temperature of 70° F. The measured pressure drop across the filter (corrected to standard conditions of 70° F and 29.92 inches of mercury total pressure) shall be no greater than that specified in 3.3. The airflow measurement in scfm, the air temperature at the inlet of the filter, the pressure differential across the filter, and the ambient barometric pressure shall be recorded as the basis for determination of compliance. Correction for non–standard conditions is not required when it is clear that the passage or failure of the filter is not in doubt. (See note 6.6)."

4.4.6.2: Add a new paragraph 4.4.6.2 as follows:
"4.4.6.2 Filtration efficiency. The test apparatus used to measure the filtration efficiency shall be capable of maintaining a stable concentration of aerosol with a mean particle size of 0.3 microns with a steady flow through the filter of 20 scfm. The test apparatus shall have the capability to determine the concentration of aerosol both upstream and downstream of the filter within 0.005%. The downstream sample point shall be located such that changing of its location across the ductwork does not cause a significant change in measured concentrations. This shall be verified using a filter with a known measurable leak. The filter shall be placed

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in the test apparatus such that the flow of air containing the challenge aerosol is in the direction indicated by the marking on the filter, and the flow of aerosolized air shall be maintained at 20± 3 scfm. This flow shall be maintained for not less than 1 minute or more than 2 minutes. The filter shall be rejected if the downstream concentration exceeds 0.03 percent of the upstream concentration which indicates that the filter efficiency is less than 99.97 percent. (See note 6.8.)"

Renumber paragraphs under 4.4.6 accordingly.

4.4.6.2, last sentence:
Delete "...resistance to ...4.4.6.3."and substitute "...resistance to airflow and filter efficiency in accordance with 4.4.6.1 and 4.4.6.2 and tested for air leakage in accordance with 4.4.6.4."

4.4.6.3:
Line 7: Delete "...15± 1 cubic centimeters..."and substitute "...0.91± 0.06 cubic inches..."
Last line: Delete "...15 cubic centimeters..."and substitute "...0.91 cubic inches..."

4.4.6.3, line 5: Delete ".032 inch"and substitute ".120 inch"

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6.5: Delete "Dioctylphthalate (DOP)". Following 6.5, add a new paragraph:

"6.6 When flows are measured in scfm (mass flow), the measurements are biased by variations from standard conditions which may exist within the test environment, higher than standard temperatures and lower than standard conditions of 70° F and 29.92 inches of mercury. While temperature can be reasonably controlled, barometric pressure varies naturally with time and location. These effects may be corrected by scientifically accepted computational methods. Calculation of corrections is not needed when the test is performed under conditions of higher than 70° F and lower than 29.92 inches of mercury total pressure when the raw test result indicates passage, or when the test is performed under conditions lower than 70° F and higher than 29.92 inches of mercury total pressure when the raw test result indicates failure; since, acceptance or rejection of the filter could not be changed by the result of such corrections."

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Add a new paragraph as follows:

"6.7 Filtration efficiency tests and aerosol materials. The TDA 100 Filter Penetration Tester manufactured by Air Techniques Inc. has been found to be capable of performing the filtration efficiency test when operated with Poly–Alpha Olefin (common trade names are Ethyl Flow 164 and Emery 3004) or Dioctylphthalate (DOP) as the aerosol agent. DOP is not recommended because of its potential as a carcinogen. Filtration efficiency is calculated by subtracting the measured percent penetration from 100 percent.

%P = [(PPM downstream)/(PPM upstream) x 100%
Eff = 100% – (%P)

Where:
PPM – Particles per minute
%P – Percent penetration
Eff – Filtration efficiency"

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